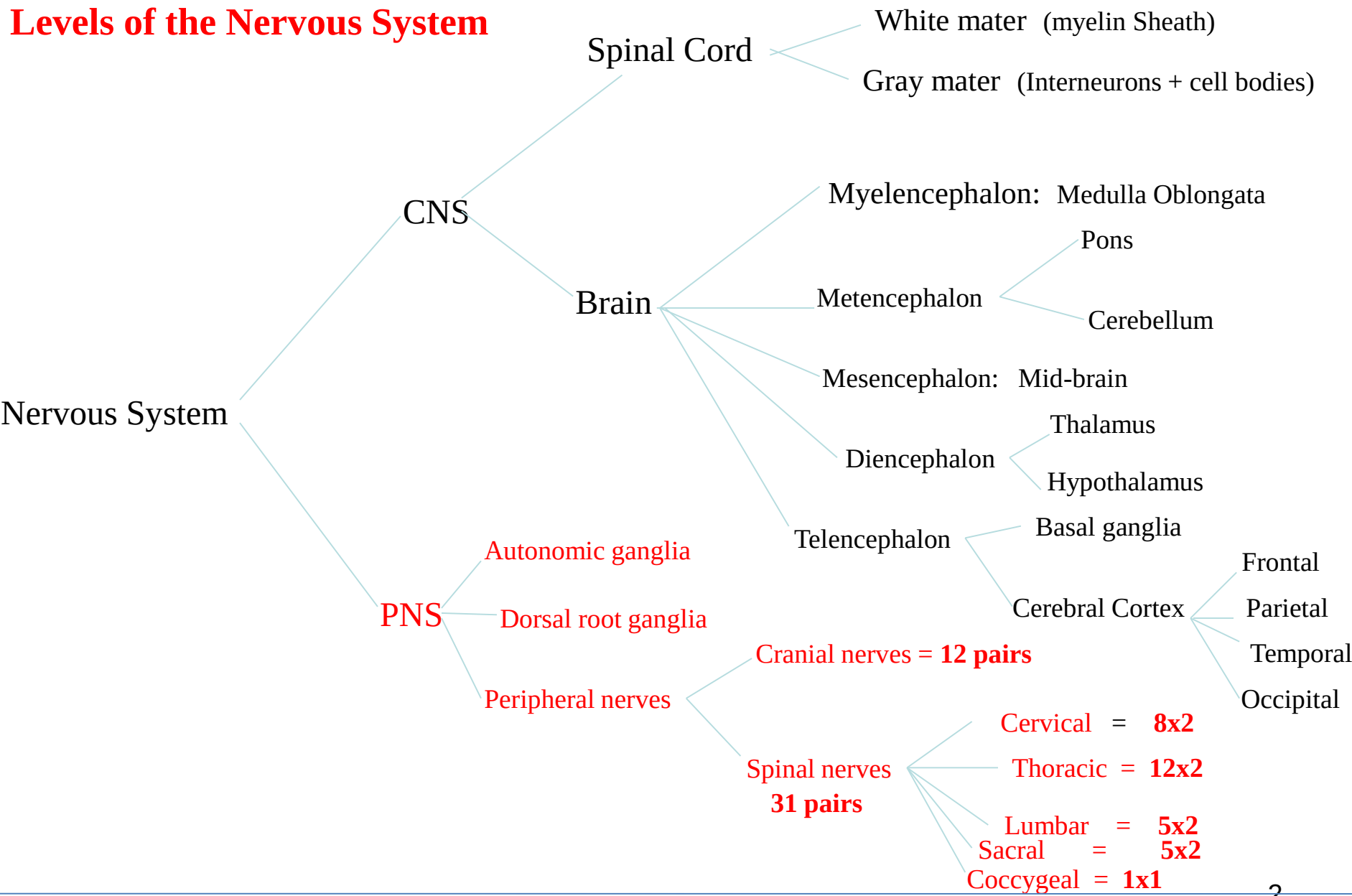
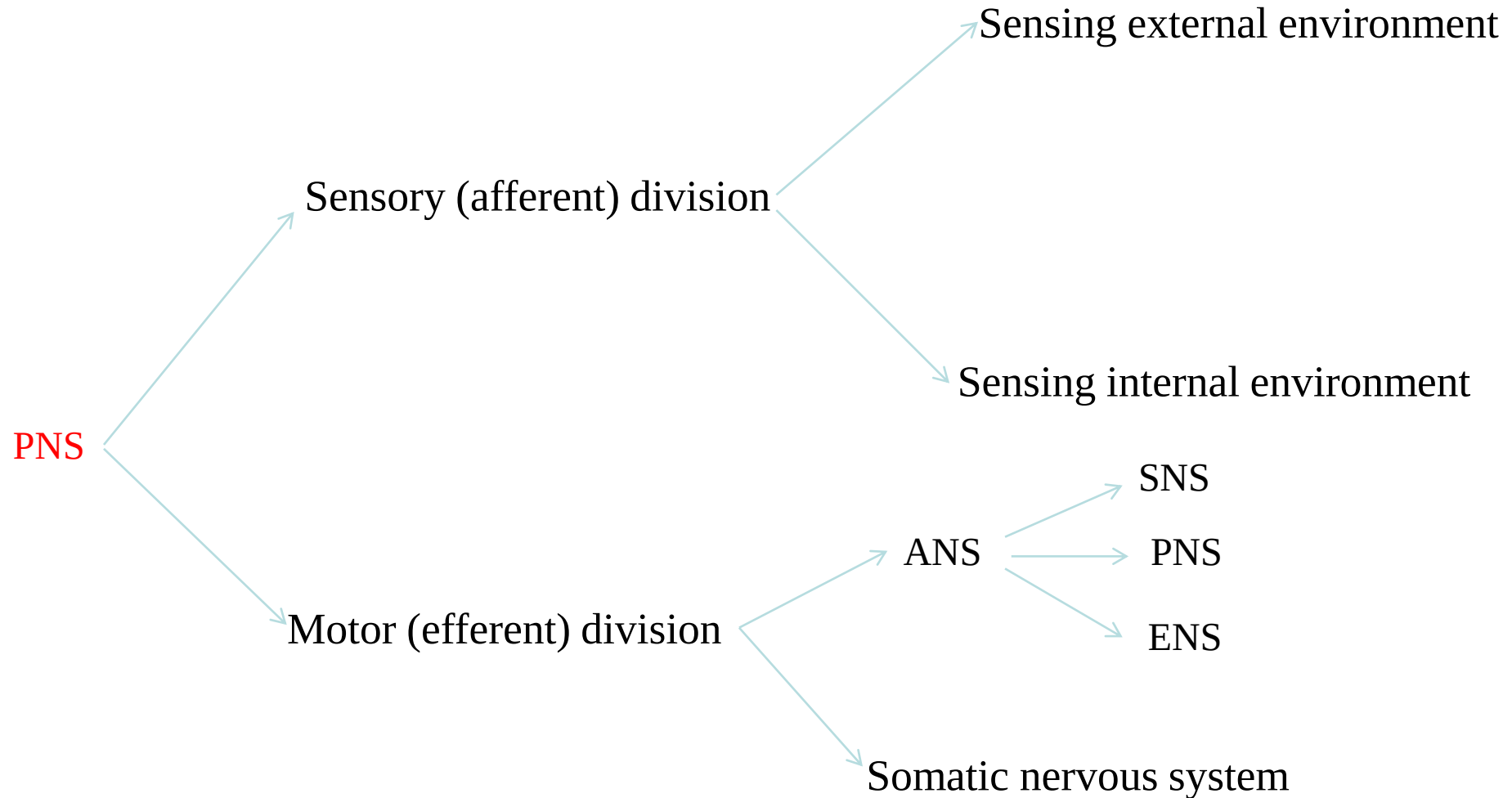


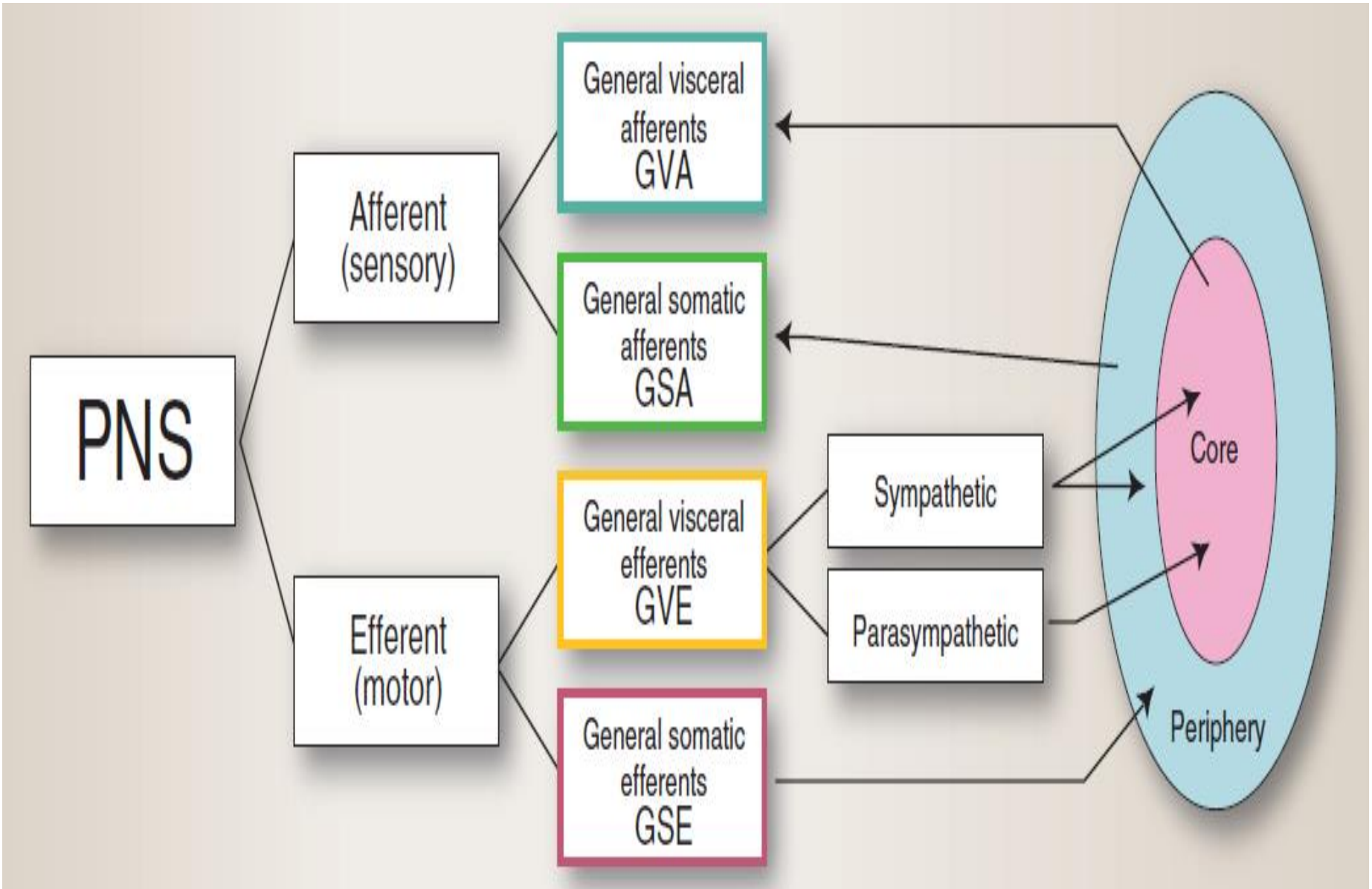
The Autonomic Nervous System

Levels of the Nervous System





+++ Somatic + visceral receptors



Peripheral Nervous System

- Somatic nervous system
 - Generally voluntary
 - Conduct impulses from the CNS to skeletal muscles
- Autonomic nervous system
 - Generally involuntary
 - Conducts impulses from the CNS to smooth muscle, cardiac muscle, and glands

Somatic vs Autonomic NS

Somatic NS

1. Control voluntary muscle (skeletal muscle)
2. Nerve fibres are originated from the anterior horn of the grey matter of the SC.
3. The motor nerve contains single, long, thick and myelinated axon.
4. The NT is always Ach and the receptor is always NR

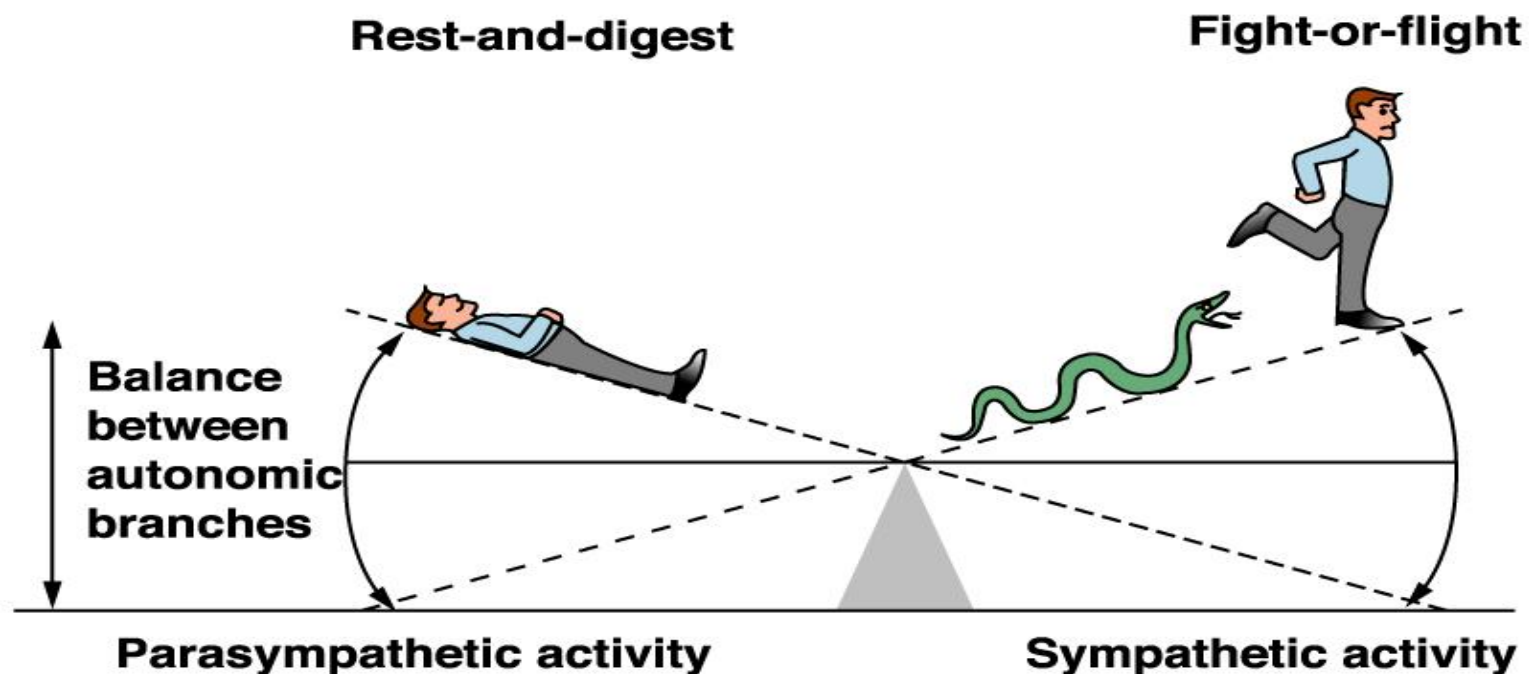
Autonomic NS

1. Control involuntary muscles (smooth and cardiac muscle)
2. Nerve fibres are originated from the lateral horn of the grey matter of spinal cord.
3. Autonomic fibres contain two neurons
4. There are both cholinergic and adrenergic fibres acting on cholinergic and adrenergic receptors

Autonomic Nervous System

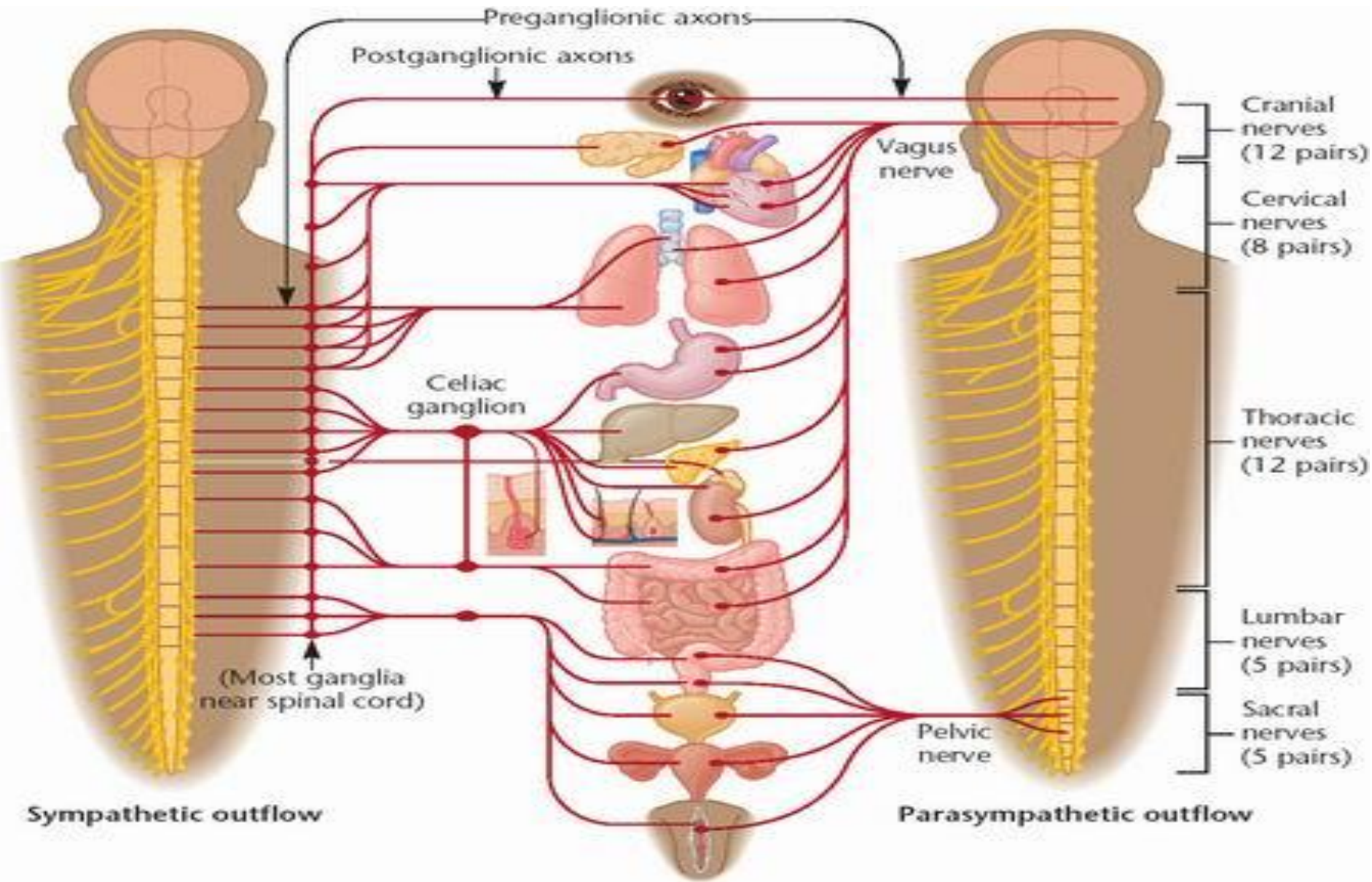
- **Division:**

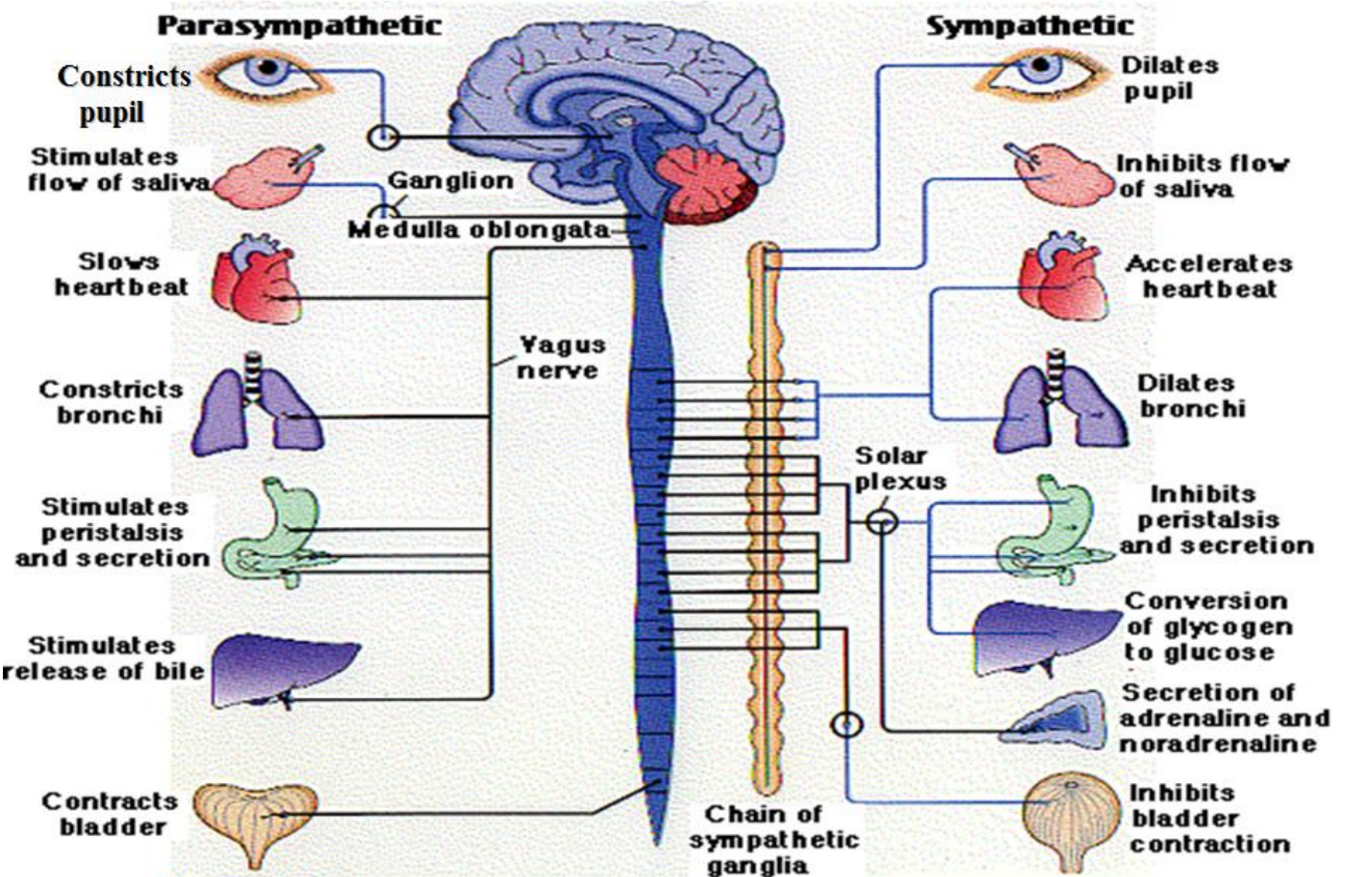
- Sympathetic Nervous System → “Fight or Flight”
- Parasympathetic Nervous System → “Rest and Digest”



Autonomic Nervous System

- **Dual innervation:**
 - Most body structures receive dual innervations of SNS and PNS
 - Purpose → better control
- The following structures are solely innervated by the SNS:
 - Sweat glands
 - Adrenal medulla
 - Blood vessels
 - Skin pilo erectors





Autonomic Nervous System

- Thoracolumbar out flow:
 - Origin → lateral horns of the spinal cord (T1-T12 and L1-L2)
 - Contain short preganglionic neurons (Cholinergic)
 - Cell bodies are located in the thoracic and lumbar spinal cord
 - Preganglionic axons synapse in lateral/collateral ganglia, located near the spinal cord far away from the organs being innervated
 - Long Postganglionic axons (Cholinergic & Adrenergic)
 - Travel from the lateral/collateral ganglia to the target organ

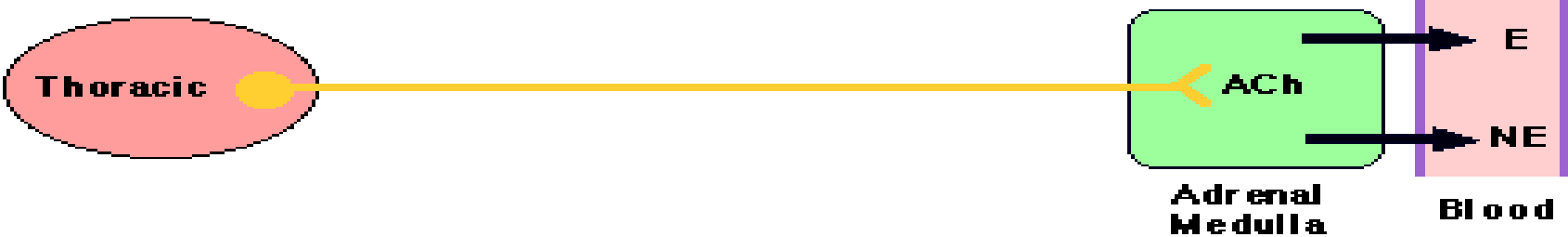
Parasympathetic:



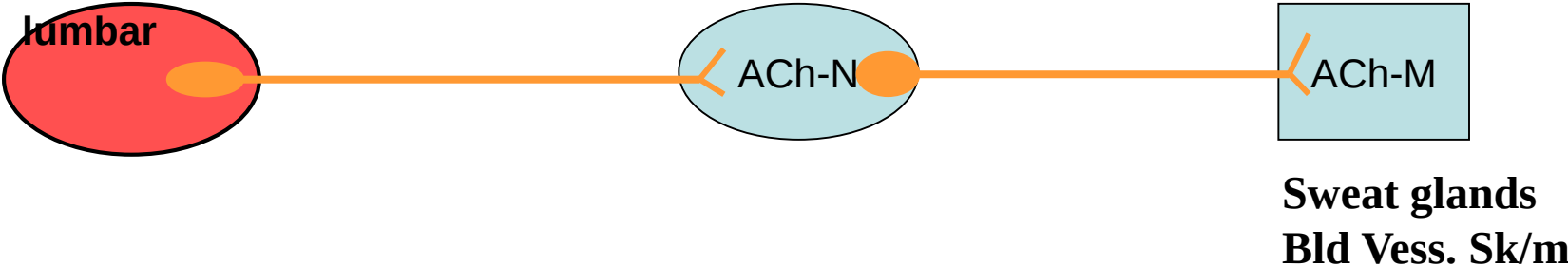
Sympathetic:



Adrenal Medulla:



Thoraco lumbar



Autonomic Nervous System

- Thoracolumbar out flow: function
 - Has the property of adjusting in response to stressful situations (trauma, fear, hypoglycemia, cold, exercise...)
 - Continually active to some degree (in maintaining the tone of vascular beds)
 - Usually concerned with expenditure of energy → Increases BMR
 - Works with adrenal medulla → epinephrine
 - Increases CO during exercise and excitements
 - Sympathetic system is not essential to life

Autonomic Nervous System

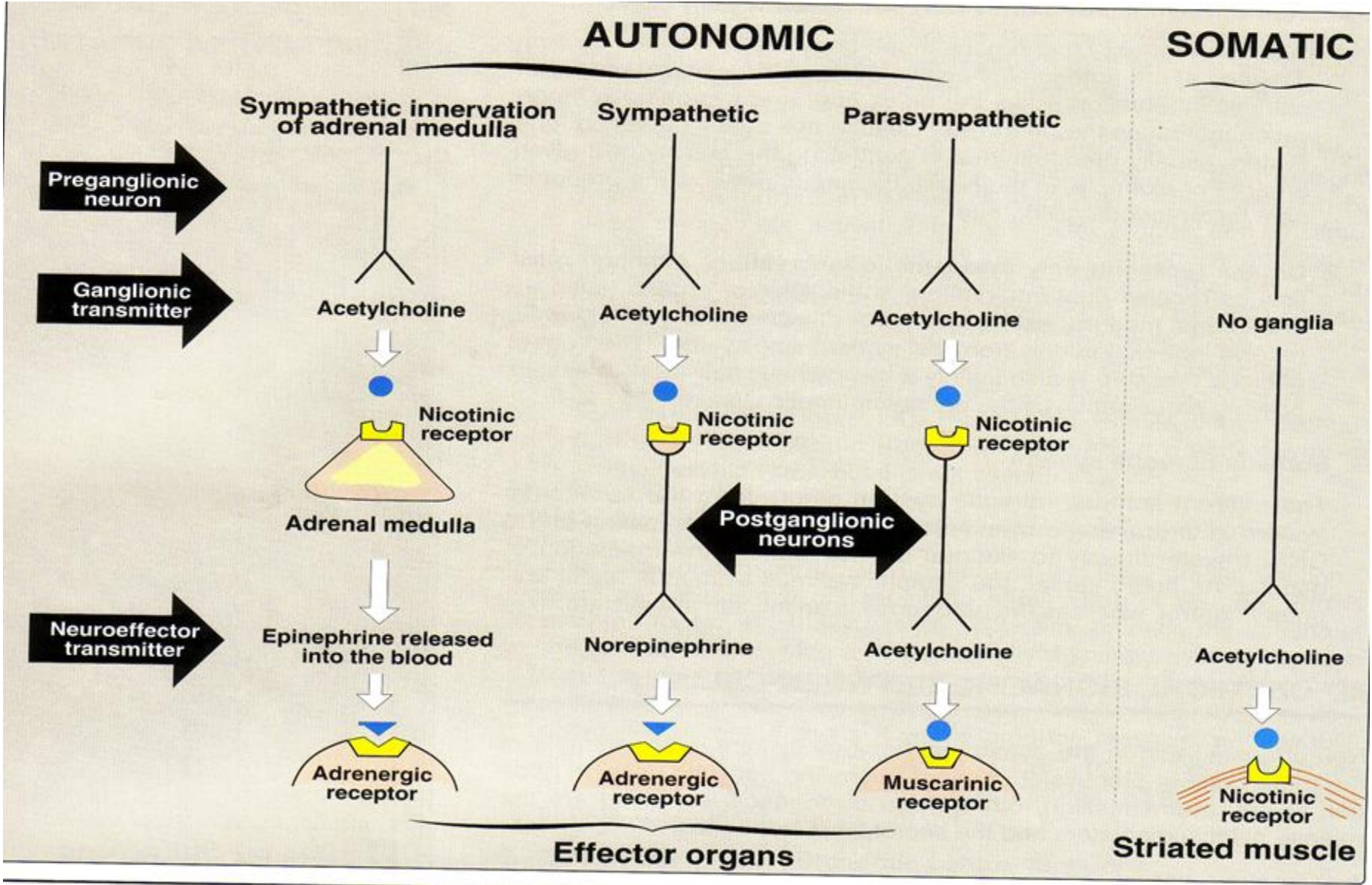
- Craniosacral out flow:
 - Origin → Cranial (C-III, VII, IX, X) and Sacral (S2-S4) segments
 - Contain long preganglionic neurons (Cholinergic)
 - Cell bodies are located in parts of certain nuclei of cranial nerves and in the sacral part of the spinal cord
 - Preganglionic axons synapse in terminal ganglia which are located close to or on the organ being innervated
 - Short postganglionic axons (Cholinergic)
 - Travel from the terminal ganglia to the target organ

Autonomic Nervous System

- **Craniosacral out flow: function**
 - Generally inhibitory effect on almost all body tissues except in the GIT (excitatory):
 - Regulate digestion, defecation and micturition
 - Conservation of energy (anabolic) → maintains a homeostatic environment "resting and digesting" system
 - No mass discharge, discrete activities
 - Tonic discharge to the heart

Autonomic Nervous System

- **Characteristics:**
 - All preganglionic neurons are cholinergic neurons
 - Secrete acetylcholine (ACh) and are excitatory
 - Parasympathetic post ganglionic neurons are cholinergic
 - Secrete ACh and are either excitatory or inhibitory
 - Most Sympathetic post ganglionic neurons are adrenergic:
 - Release NE and are either excitatory or inhibitory
 - Sympathetic cholinergic post ganglionic neurons innervate:
 - Sweat glands, pilo-erector muscles, adrenal medulla



Autonomic Nervous System

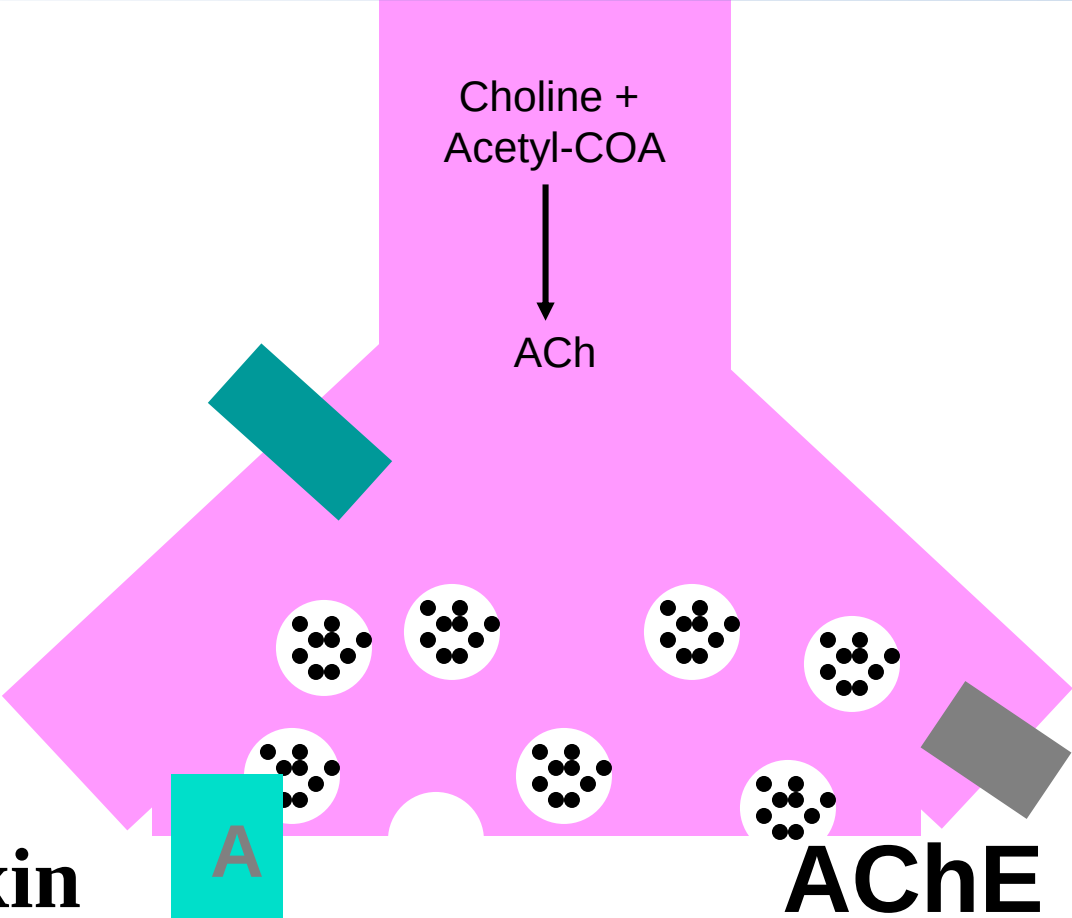
- Autonomic neurotransmitters:
 - Acetylcholine (ACh)
 - Norepinephrine (NE)
- Released by the presynaptic neuron
- Bind to specific receptors in the postsynaptic cell membrane
 - Binding has either an excitatory or an inhibitory effect on the effector, depending on the specific receptor
 - The type of receptor determines the effect of the NT

Synthesis of ACh

choline
acetyltransferase



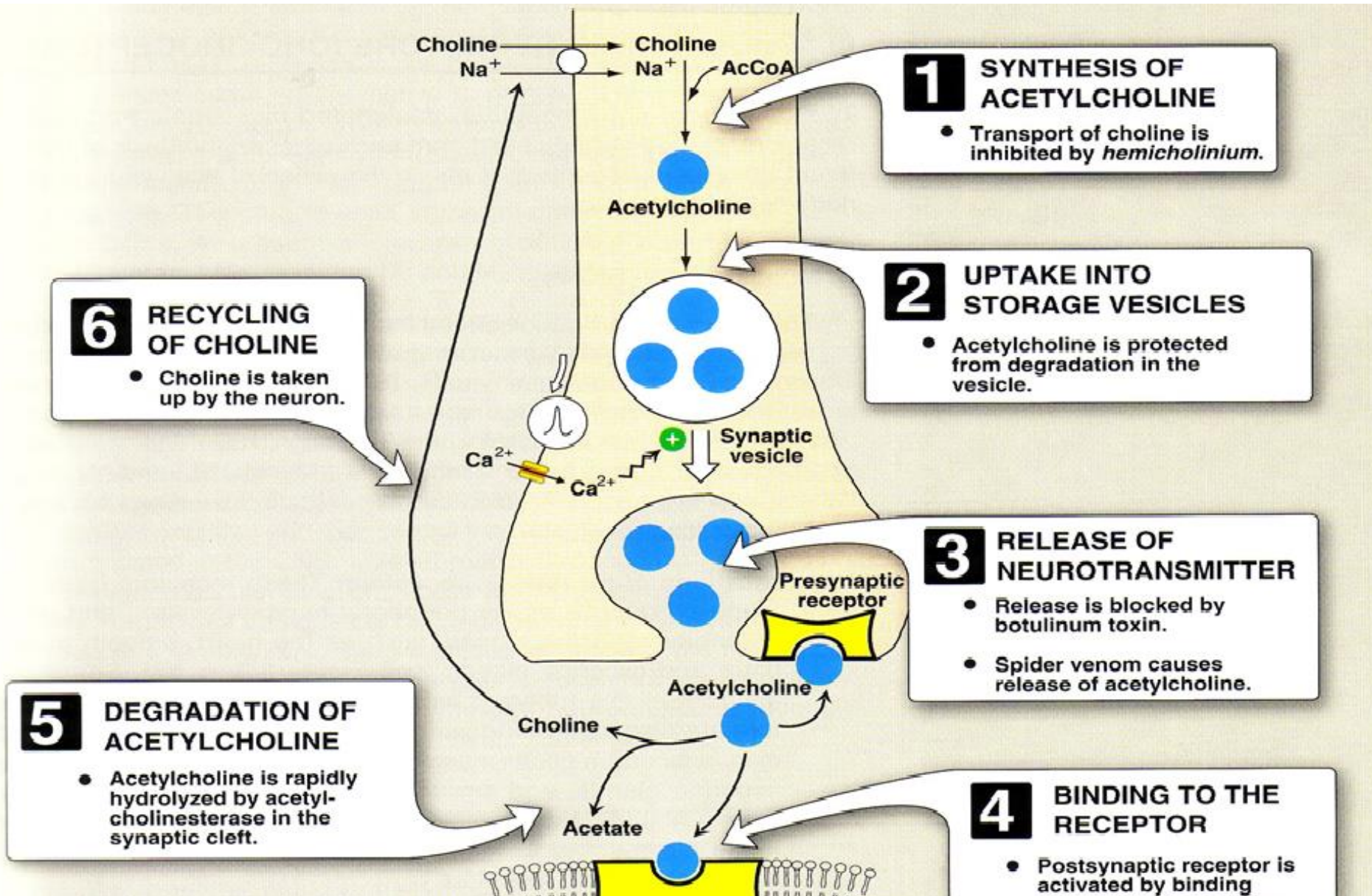
ACh



Botulinum toxin

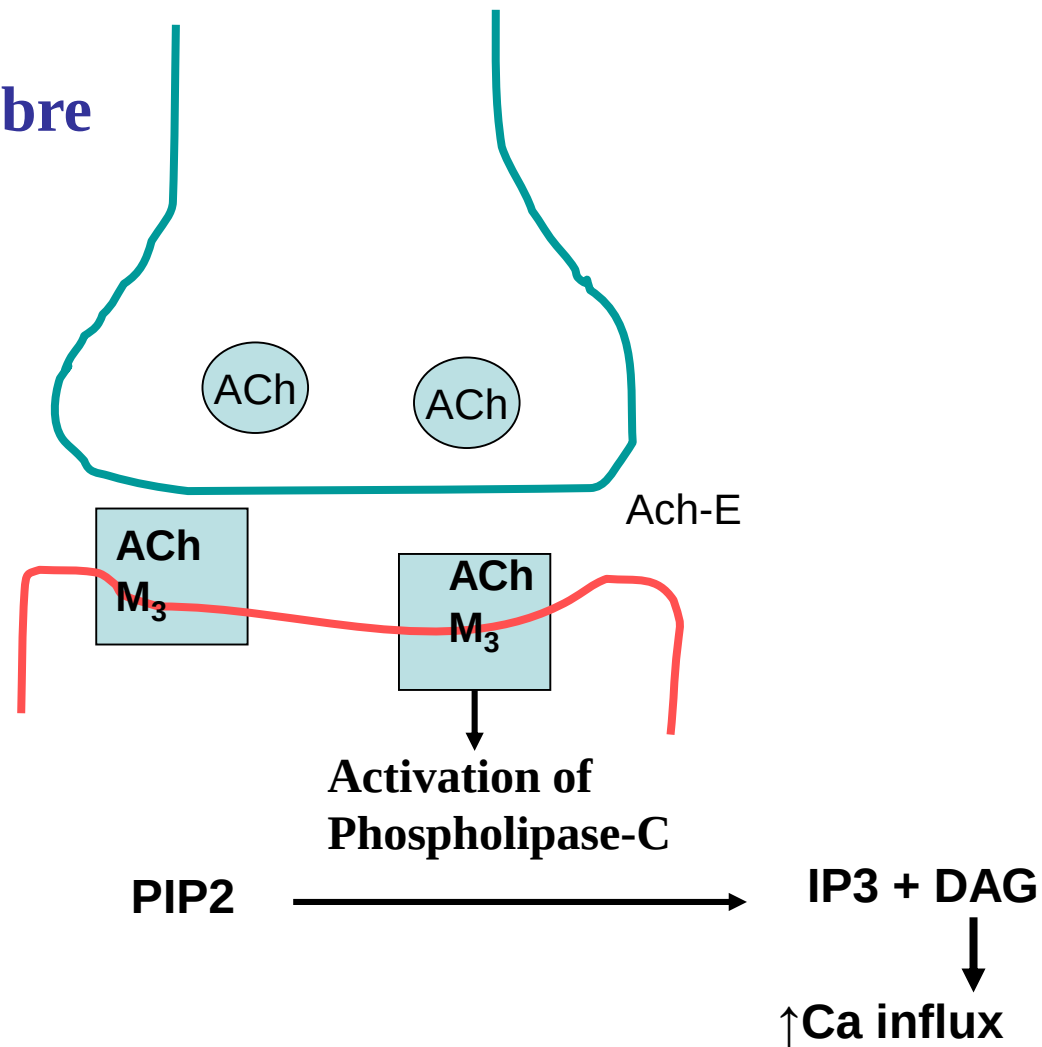
Inhibit release



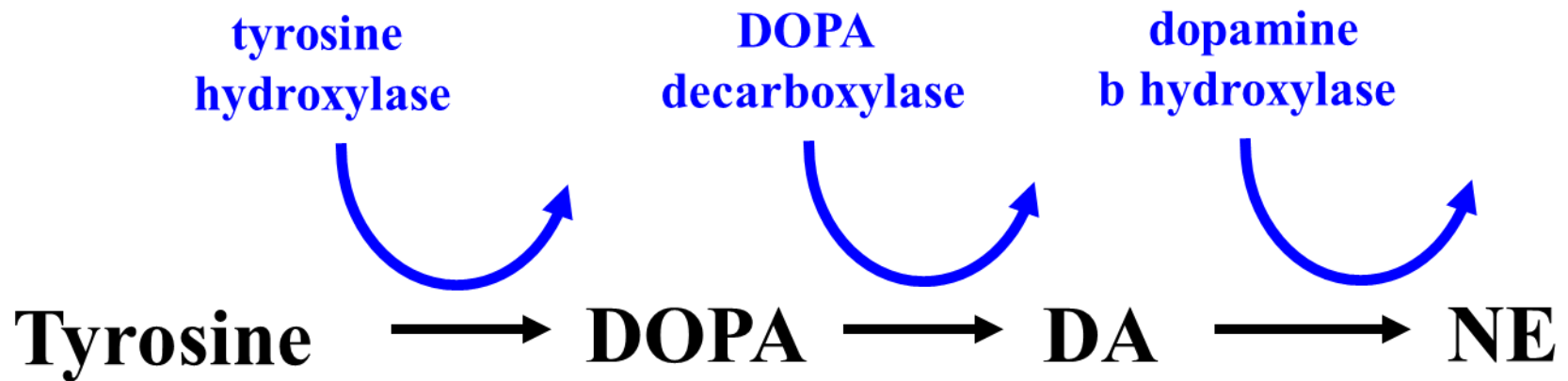


Mechanism of action of ACh

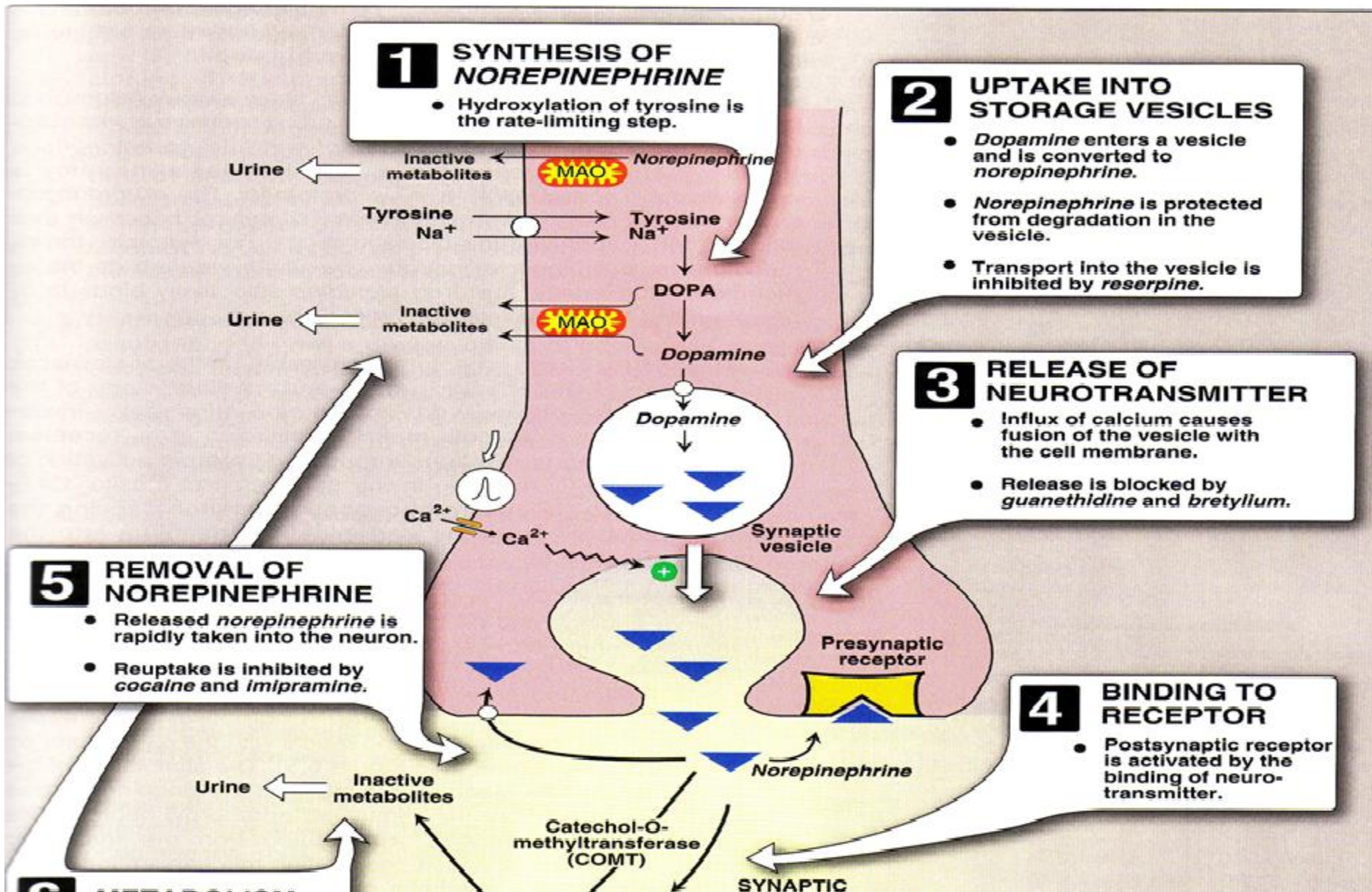
A Cholinergic fibre

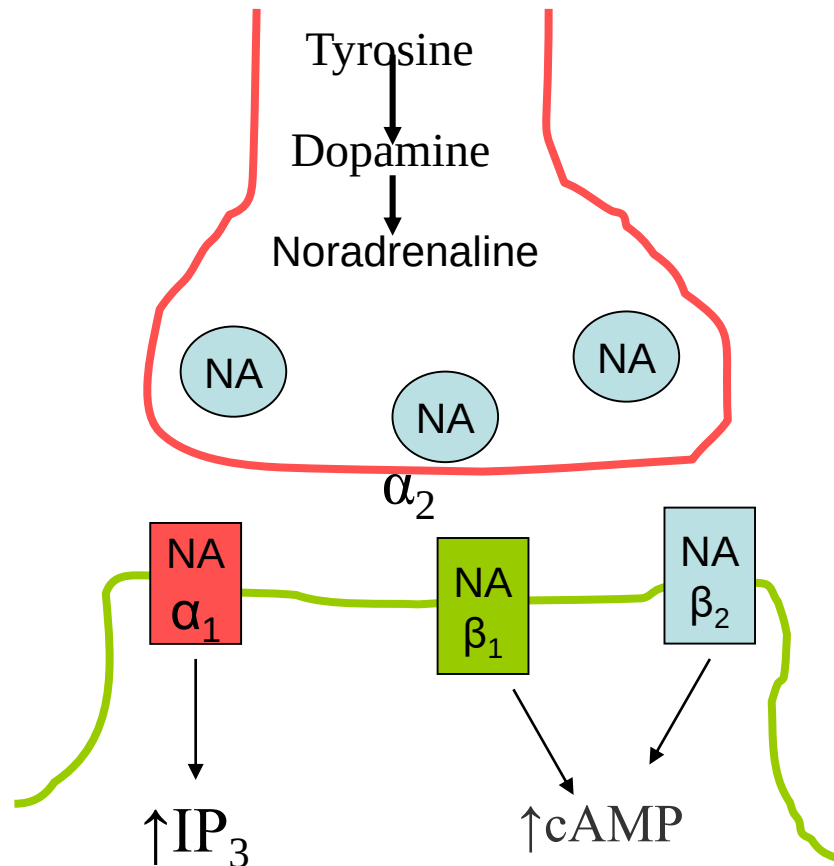


Synthesis of NE



- NE is metabolized by MAO and COMT





Mechanism of action of Catecholamines

Autonomic Nervous System

- Cholinergic Receptors:
 1. Nicotinic receptors (N)
 - Found between pre/post ganglionic neurons of both PNS & SNS (Nn) + at the neuromuscular junction (NMJ, Nm)
 2. Muscarinic receptors (M1-M5)
 - Found on all effector cells + stimulated by both PSN and SNS
- Adrenergic Receptors:
 1. Alpha receptors $\rightarrow \alpha$ -1 and α -2
 2. Beta receptors $\rightarrow \beta$ -1, β -2 and β -3

Cholinoceptors	Typical Locations	Result of Ligand Binding
Muscarinic M ₁	CNS neurons, sympathetic postganglionic neurons	Formation of IP ₃ and DAG, increased intracellular Ca ²⁺
Muscarinic M ₂	Myocardium, smooth muscle, CNS neurons	Opening of K ⁺ channels, inhibition of adenylyl cyclase
Muscarinic M ₃	Exocrine glands, vessels (smooth muscle and endothelium); CNS neurons	Like M ₁ receptor-ligand binding
Muscarinic M ₄	CNS neurons; possibly vagal nerve endings	Like M ₂ receptor-ligand binding
Muscarinic M ₅	Vascular endothelium, especially cerebral vessels; CNS neurons	Like M ₁ receptor-ligand binding
Nicotinic N _N	Postganglionic neurons, some presynaptic cholinergic terminals	Opening of Na ⁺ , K ⁺ channels, depolarization
Nicotinic N _M	Skeletal muscle neuromuscular endplates	Opening of Na ⁺ , K ⁺ channels, depolarization

Receptor	Location	Mechanism	Major Functions
M ₁	Nerve endings	G _q -coupled	↑ IP ₃ , DAG cascade
M ₂	Heart, some nerve endings	G _i -coupled	↓ cAMP, activates K channels
M ₃	Effector cells: smooth muscle, glands, endothelium	G _q -coupled	↑ IP ₃ , DAG cascade
N _N	ANS ganglia	Ion channel	Depolarizes, evokes action potential
N _M	Neuromuscular end plate	Ion channel	Depolarizes, evokes action potential

Adrenoceptors	Typical Locations	Result of Ligand Binding
Alpha ₁	Postsynaptic effector cells, especially smooth muscle	Formation of IP ₃ and DAG, increased intracellular Ca ²⁺
Alpha ₂	Presynaptic adrenergic nerve terminals, platelets, lipocytes, smooth muscle	Inhibition of adenylyl cyclase, decreased cAMP
Beta ₁	Postsynaptic effector cells, especially heart, brain, Juxta-glomerular cells	Stimulation of adenylyl cyclase, increased cAMP
Beta ₂	Postsynaptic effector cells, especially smooth muscle and cardiac muscle	Stimulation of adenylyl cyclase and increased cAMP.
Beta ₃	Postsynaptic effector cells, especially lipocytes, ...	Stimulation of adenylyl cyclase and increased cAMP ¹

Receptor	Location	G Protein	Second Messenger	Major Functions
α_1	Effector tissues: smooth muscle, glands	G_q	$\uparrow IP_3$, DAG	$\uparrow Ca^{2+}$, causes contraction, secretion
α_2	Nerve endings, some smooth muscle	G_i	$\downarrow cAMP$	$\downarrow$ Transmitter release, causes contraction
β_1	Cardiac muscle, juxtaglomerular apparatus	G_s	$\uparrow cAMP$	$\uparrow$ Heart rate, $\uparrow$ force; $\uparrow$ renin release
β_2	Smooth muscle, cardiac muscle	G_s	$\uparrow cAMP$	Relax smooth muscle; $\uparrow$ glycogenolysis; $\uparrow$ heart rate, force
β_3	Adipose cells	G_s	$\uparrow cAMP$	$\uparrow$ Lipolysis
D_1	Smooth muscle	G_s	$\uparrow cAMP$	Relax renal vascular smooth

Organ	Effect of Sympathetic Stimulation	Effect of Parasympathetic Stimulation
Eye		
Pupil: Iris radial muscles (α_1 -AR)	Contraction leads to pupillary Dilation (Mydriasis)	None
Pupil: Iris circular muscles (M-R)	None	Contraction leads to pupillary constriction (Miosis)
Ciliary muscle	Slight relaxation (far vision)	Constricted (near vision)
Glands	Vasoconstriction and slight secretion	Stimulation of copious secretion
Nasal	+	+ +
Lacrimal	+	+ +
Salivary glands	+	+ +
GIT glands	Inhibited	+ + +
Pancreatic secretions	Inhibited	Increased
Sweat glands	Copious sweating (cholinergic)	Sweating on palms of hands
Apocrine glands	Thick, odoriferous secretion	None
Systemic Blood vessels	Most often constricted	Most often little or no effect

Organ	Effect of Sympathetic Stimulation	Effect of Parasympathetic Stimulation
Heart	Enhances all activities	Inhibits all activities
Muscle	+ve Chronotropic effect (↑ HR)	Slow HR
	+ Inotropic effect (↑ Force of contraction)	Decreased force of contraction (esp. of atria)
Coronary vessels	Vasodilation (β_2)	Vasoconstriction
Lungs		
Bronchi	Bronchodilation (β_2)	Bronchoconstriction (M_3)
Blood vessels	Mildly constricted	? Dilated
Gut	Inhibits all activities	Enhances all activities
Lumen	Decreased peristalsis and tone	Increased peristalsis and tone
Sphincter	Increased tone (most times)	Relaxed (most times)

Organ	Effect of Sympathetic Stimulation	Effect of Parasympathetic Stimulation
Liver	↑Glycogenolysis ↑Glucose released	Slight glycogen synthesis
Gallbladder and bile ducts	Relaxed	Contracted
Kidney	Vasoconstriction ↓RBF = ↓UO and ↑renin secretion	None
Bladder		
Detrusor	Relaxed (slight)	Contracted
Trigone	Contracted	Relaxed
Penis	Ejaculation	Erection
Systemic arterioles		
Abdominal viscera	Constricted	None
Muscle	Constricted (α_1)	None
	Dilated (β_2)	
	Dilated (cholinergic)	
Skin	Constricted	None

Organ	Effect of Sympathetic Stimulation	Effect of Parasympathetic Stimulation
Blood		
Coagulation	Increased	None
Glucose	Increased	None
Lipids	Increased	None
Basal metabolism	Increased up to 100%	None
Adrenal medullary secretion	Increased	None
Mental activity	Increased	None
Piloerector muscles	Contracted	None
Skeletal muscle	Increased glycogenolysis	None
	Increased strength	None
Fat cells	Lipolysis	None

Organ or Tissue	Effect of Sympathetic Discharge (Receptor Type)	Effect of Parasympathetic Discharge (Receptor Type)
Eye	Dilates pupil (α)	Constricts pupil, focuses for near vision (M)
Airways	Dilates bronchioles (β_2)	Constricts bronchioles (M)
GI tract	Slows motility (α , β_2)	Increases motility, secretion (M)
GU tract	Contracts sphincters, mediates ejaculation (α)	Contracts walls of bladder (M), mediates erection
Vessels	Constricts arterioles in skin and splanchnic vessels (α), dilates in skeletal muscle vessels (β_2)	Little effect
Heart	Accelerates all pacemakers and AV conduction, increases force of contraction (β_1 , β_2)	Slows sinus rate and AV conduction, increases AV refractory period (M)
Exocrine glands	Increases sweating (M), salivation (slight, α)	Increases salivation markedly, increases lacrimal, gastric, duodenal, and pancreatic secretion (M)

Autonomic Reflexes

- Reflex → fast and involuntary response to a stimulus
- Consists of an action signaled to the CNS and a reaction by the CNS
- Any reflex is transmitted through a reflex arc
- A **reflex arc** has 5-components:
 1. Receptor → detect changes
 2. Afferent (sensory) pathway
 3. Integrating centre (spinal cord, brain)
 4. Efferent (motor) pathway
 5. Effector organs → muscles and secretory glands

Autonomic Reflexes...

- Representative autonomic reflexes:
 - The Baroreceptor reflex
 - The chemoreceptor reflex
 - Defecation reflex
 - Micturition reflex